

Fahim Ahamed

Authorized to work in the U.S. and do not require sponsorship - U.S. Permanent Resident

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Summary

Data Scientist with a strong background in **AI Research**, with experience across **healthcare**, **NLP**, and **geoscience** domains. Builds **end-to-end ML pipelines**, conducts **statistical and spatial analysis**, and deploys models that hold up in production. Develops **agentic AI pipelines** using **RAG**, **LLMs**, and **MCP**. Published across **IEEE**, **SAGE**, and **Springer Nature**. Comfortable working across the full stack from raw data to results, and collaborating directly with domain experts. Proficient in **Python**, **SQL**, **R**, **PyTorch**, and **TensorFlow**.

Work Experience

Research Assistant

Dec 2025 - Present

CUNY Research Foundation

New York, NY

- Developing an **agentic AI pipeline** using **RAG**, **LLMs**, and **MCP-connected tools** to enable natural language querying over complex geoscience databases.
- Bridging the gap between **geoscience domain expertise** and **AI engineering**, translating research requirements into technical specifications and iterating on pipeline outputs to ensure interpretability and scientific relevance.
- Designing the **end-to-end technical stack** spanning **vector database architecture**, domain-specific **embedding models**, **LLM optimization**, and **cloud deployment** to support a production-ready geoscience research tool.

Data Evangelist Intern

Feb 2026 - May 2026

DataCamp

New York, NY

- Designed and maintained **scalable, reproducible pipelines** to collect, integrate, and validate **multi-source data** using **APIs** and **web scraping**.
- Built and published **data applications** and **analytical workflows**, transforming raw data into **user-facing tools** for exploring key metrics.
- Analyzed data to identify **trends, patterns**, and **key drivers**, producing clear outputs that support **data-driven decision-making**.

Data Analyst

Oct 2025 - Feb 2026

The City College of New York

New York, NY

- Led analysis of **institutional data** using **Excel**, **Tableau**, and **Python**, surfacing patterns that shaped **advising and student engagement programs**.
- Conducted **data-driven surveys** and **retention studies**, uncovering behavioral patterns that inform targeted outreach.
- Built **analytical reports** and **dashboards** that surfaced trends and anomalies across departments, replacing gut-feel decisions with **data-backed ones**.

Technical Skills

Languages: Python, SQL, R

Data Analysis: NumPy, Pandas, SciPy, Excel

Data Visualization: Matplotlib, Seaborn, Plotly, ggplot2, Tableau

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, NLTK, Word2Vec

LLMs & Agents: LLMs, RAG, MCP, Hugging Face, ChromaDB

Database & Big Data: MySQL, MongoDB, Snowflake

MLOps & Cloud: AWS, MLflow, Git/GitHub, Docker, Vercel

Others: Data Preprocessing, Feature Engineering, Model Evaluation, A/B Testing

Education

The City College of New York

New York, NY

M.S. in Data Science and Engineering

May 2027

East West University

Dhaka, Bangladesh

B.S. in Computer Science and Engineering

Dec 2023

Research Experience

Explainable AI-Driven Hybrid Transfer Learning Framework for Accurate Skin Cancer Diagnosis – *Digital Health (SAGE)*

- Achieved **98.65% accuracy (F1 98.64%)** on multi-class skin cancer classification using a hybrid EfficientNetB0 + Random Forest model with advanced preprocessing, validated across datasets.

Robust Dual-Site Cancer Screening via Multi-Scale Vision Transformer and Rapid Recognition Pipeline – *IEEE Xplore*

- Developed a **multi-scale Vision Transformer framework** (Multi-ViT, MA-Transformer V2, MobileUNETR, TinyViT, Xception) achieving **99.12% accuracy** on PAD-UFES-20 and **99.37%** on Oral Cancer datasets, with strong F1 scores.

Texture Feature-Based Colonic Polyp Detection Using Deep Learning Techniques – *Springer Nature*

- **Achieved 97.33% IoU** for colonic polyp segmentation using DeepLabv3+, outperforming VGG16 and prior studies to set a new benchmark in colorectal cancer detection.

Optimizing Energy Usage: Genetic Algorithms Leading the Charge Toward Sustainability – *Springer Nature*

- **Reduced** annual household electricity usage by **34.54%** (90,449 watts) using **Genetic Algorithms**, demonstrating strong environmental impact and potential for **nationwide scalability** in sustainable energy optimization.

Accepted Papers (In Press)

A Lightweight Transformer Ensemble for Explainable Depression Emotion and Severity Detection – *IEEE (2025)*

- Achieved **state-of-the-art F1 81.63%, Precision 83.24%, Specificity 86.92%** on benchmark datasets with a lightweight transformer ensemble, providing **LIME-based explanations** and a deployable real-time interface for depression detection.

Papers Under Review

Multimodal Learning in Medical Imaging: Methods, Benchmarks, Evaluation, & Clinical Translation – *Elsevier*

- Conducted a structured review of **multimodal learning in medical imaging**, proposing a unified taxonomy while identifying key gaps in **robustness, reproducibility, and benchmark standardization** across healthcare AI systems.

Explainable Token-Fusion Transformer for Early Drowsiness State Recognition in Safety-Critical Systems – *Array Journal*

- Designed **EMTF-ViT**, achieving **99.2% intra-dataset accuracy** and **93%+ cross-dataset transfer**, with Grad-CAM interpretability and real-time low-power deployment for ADAS and safety applications.

Data Projects

NYC HIV/AIDS Trend Analysis & Predictive Modeling | *R, Tidyverse, ggplot2, sf, rpart, RandomForest, caret*

- Cleaned and engineered **multi-year NYC Open Data** on HIV/AIDS diagnoses, resolving suppressed values, harmonizing demographic fields, and producing analysis-ready datasets for modeling.
- Performed **spatiotemporal analysis of HIV/AIDS trends across NYC**, uncovering persistent geographic and demographic disparities, and leveraged **Random Forest models ($R^2 = 0.65$)** to identify high-impact predictors for targeted public health interventions.

Global Population Growth Modeling & Forecasting | *R, Tidyverse, ggplot2, leaps, caret*

- Engineered a **longitudinal time-series dataset** from World Bank indicators, integrating population, fertility, mortality, migration, and age structure data across **200+ countries**, and conducted **multi-decade demographic analysis** by region and income group.
- Developed and validated **population growth forecasting models**, achieving **$R^2 = 0.755$** , and applied **rolling-window cross-validation** to assess short- and long-horizon predictive stability.

AI/ML Projects

Clinical Risk Modeling & Unsupervised Structure Analysis | *Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*

- Curated and standardized **clinical tabular data**, addressing missingness, outliers, and feature scaling to support supervised prediction and unsupervised structure discovery.
- Performed **diagnostic exploratory analysis** to identify risk-associated variables, then developed a **logistic regression classifier (80% test accuracy)** and evaluated **K-Means clustering with PCA and t-SNE projections** to assess latent structure in medical risk profiling.

Intrusion Detection System | *Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*

- **Processed and analyzed 2.5+ million network traffic records** through comprehensive EDA and reduced memory usage by **47.5%** through numerical datatype downcasting, improving training efficiency.
- **Achieved 97% median accuracy** in detecting network anomalies using machine learning models, uncovering key traffic patterns and strengthening anomaly detection capabilities.

Additional

- Solved **500+ problems** on platforms like **LeetCode, HackerRank, and Beecrowd** (formerly URI), combined.